Fiscal Year:	FY 2013	**Task Last Updated:**	FY 07/19/2012
PI Name:	Law, Chung K. Ph.D.		
Project Title:	Structure and Response of Spherical Diffusion Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	Chung Law's email is cklaw@princeton.edu		
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2009	**End Date:**	09/30/2014
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: Project to be extended to 9/30/2014 per D. Stocker/GRC (Ed., 7/20/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Tse, Stephen (Rutgers University)		
Grant/Contract No.:	NNX09AR03A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The effort was divided into experimental and theoretical elements. The experimental effort consisted of using a porous spherical burner to issue fuel mixtures into an oxidizing environment. Different mixtures of fuel species permitted investigation of the phenomena of interest. The experiment effort was conducted primarily in the drop towers at the GRC and the NASA reduced-gravity aircraft. Space flight experiment was fully defined and approved during the program period. The theoretical effort consisted of the development of computational codes simulating the combustion phenomena of interest, and interpreting the calculated results on the basis of fundamental flame physics. The purpose of this flight definition task is to gain fundamental understanding of the structure and dynamics of diffusion flames in a simple, well-defined flow field. Using a spherical geometry this work considers: 1) gas-phase transient processes, including fuel vapor accumulation inside the flame and far-field diffusion, and their influence on flame movement and the achievement of steady flame behavior; 2) extinction boundaries of steady or near-steady flames and the identification		

	of the associated chemistry at the global, reduced, and detailed levels; 3) the characteristics of partially-premixed flames; 4) flamefront pulsation and cellular instability especially near the extinction limits; 5) the dynamics of rotating flames; and 6) the behavior and structure of quasi-spherical flames in partial gravity environments. These studies are intended to significantly enhance the understanding of diffusion flames, especially providing capability for predicting flame behavior under varied conditions and configurations. As such, this work should contribute to engineering approaches to practical combustion systems, achieving increased energy conversion efficiencies and reductions in unwanted combustion emissions in both terrestrial and extraterrestrial environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enhanced understanding of flame-flow vortical interaction could lead to improved burner design and operation in terms of energy efficiency and pollutant emissions.
Task Progress:	During the reporting period results from the following investigations were published. 1. Spherical diffusion flames under rotation. Microgravity experiments were conducted in the 2.2-second drop tower and zero-gravity facility at NASA-GRC to gain fundamental understanding of the effects of spinning on an otherwise spherical diffusion flame. The flames were generated by injecting either a fuel or an oxidizer mixture from a porous burner to a controlled ambient of either an oxidizer or fuel mixture, respectively. Results show that the polar flame location scales with the angular velocity monotonically as where a is greater and smaller than unity for small and large spinning velocities, respectively. On the contrary, the equatorial flame location responds nonmonotonically to increasing spinning velocity: first increasing and then decreasing. The experimental observations agree well with the computational simulation where the simulated results demonstrate that the nonmonotonic response of the equatorial flame location is caused by dilution of the reactant concentration in the outwardly-directed radial flow by the product and inert that are carried by the inwardly-directed polar flow upon traversing the flame segment in the polar region. This work is reported in: “Response of spherical diffusion flames subjected to rotation: microgravity experimentation and computational simulation,” by S. W. Yoo, S. Chaudhuri, K. R. Sacksteder, P. Zhang, D. L. Zhu and C. K. Law, Combustion and Flame, Vol. 159, pp. 665-672 (2012) ; publication was listed in the February 2012 Task Book Bibliography. 2. Soret diffusion in heptane flames. The influence of Soret diffusion on the structure and response of n-heptane/air flames is investigated numerically with detailed reaction mechanism and transport. Unstretched freely-propagating planar premixed flames as well as stretched counterflow premixed and diffusion flames are studied, with emphasis on the separate and combined Soret effects of heptane and its oppositely-directed decomposition species H2 and H, as well as those of the major species O2, N2, CO2 and H2O. Results show that, in the unstretched case for which the flame temperature remains at its adiabatic value, Soret diffusion primarily affects the chemical kinetics embedded in the flame structure and the net effect is small; while in the stretched cases, its impact is mainly through those of heptane and the secondary fuel H2 in modifying the flame temperature, with substantial effects, while H2 and H also affect the chemical kinetics especially when the active reaction zone is localized. The dilution/enrichment of the reactant concentrations in the flame through the Soret diffusion of the major species O2, N2, CO2 and H2O further exert finite effects on the flame burning intensity. This work is reported in: “A comprehensive evaluation of Soret diffusion in heptane/air flames,” by Y. X. Xin, C. J. Sung and C. K. Law, Vol. 159, pp. 2345-2351 (2012). 3. Self-acceleration of expanding spherical flames induced by flamefront instability. The cells that continuously develop over the flame surface of an expanding spherical flame increase its area and thereby the global propagation rate, resulting in the possibility of self-acceleration. The present study examines whether this self-acceleration could be self-similar, and, if so, whether it could also be self-turbulizing. Extensive experiments at elevated pressures and thereby reduced laminar flame thicknesses and enhanced propensity to exhibit Darrieus-Landau instability were conducted for hydrogen/air over an extensive range of equivalence ratios. The results demonstrate the strong possibility of self-similar flame acceleration, strong influence of the system pressure, moderate influence of diffusional-thermal instability, and a corresponding moderate spread in the power-law acceleration exponent. This work is reported in: “An experimental investigation on self-acceleration of cellular spherical flames,” Fujia Wu, Grunde Jomaas and Chung K. Law, Proceeding of the Combustion Institute, in press. 4. Review on Future Fuel Options. The state of research on developing fuel options for next generation chemical propulsion is reviewed for aviation fuels and energetic fuels. For aviation fuels, the development is based on considerations of cost, energy security and climate change, with Fischer-Tropsch synthetic fuels and biofuels hold potential as alternative aviation fuels. The need for basic research to develop predictive capability for the oxidative chemistry of evolving fuels in evolving engine designs is emphasized, illustrated by the intricate reaction pathways and the enormity of the reaction mechanisms involved. Recent research activities towards achieving the goal of fuel design are discussed through the development of detailed mechanisms, reduced mechanisms, and surrogate fuels. For the development of high-energy-density propellants, advances in several classes of materials are discussed, including metallized and hypergolic propellants, and propellants with strained and functionalized molecules as well as nanoparticle addition. The impact of the recent progress in chemical synthesis, materials science and nano science on these advances is noted. This work is reported in: “Fuel options for next generation chemical propulsion,” by C. K. Law, AIAA Journal, AIAA Journal, Vol. 50, No. 1, pp. 19-36 (2012). doi: 10.2514/1.J051328 ; publication was listed in the February 2012 Task Book Bibliography. During the reporting period, ongoing work consisted of the following. 1. Experiments on spherical flames in low pressure: Complementary low-pressure experiments are being used to examine the flame structure of low-buoyancy, inverted, spherical diffusion flames, using laser-based diagnostics. The effect of buoyancy is minimized in a low-pressure environment, typically down to 0.1 atmosphere, beyond which kinetic effects would become severe. However, since the flames examined have centimeter-order diameters, buoyancy can still be significant due to the large dimension. Therefore, to further reduce buoyancy, an additional arrangement is implemented, which involves injecting the oxidizer gas into an environment of low molecular weight fuel such as

	hydrogen and methane. Buoyancy is reduced because the hot, low-density flame sphere is now situated in an environment of low molecular weight gas. As such, spherical and rotating non-buoyant diffusion flames can be investigated, for extended durations. Detailed measurements are being made of the flame configuration, as well as its temperature and species (CH and OH) profiles, for different fuels and mixtures involving hydrogen, methane, and ethylene, along with inerts composed of nitrogen, helium, and carbon dioxide. 2. Transient numerical simulation: The transient numerical simulation of spherical flames developed as part of this program includes detailed chemistry and transport models, especially radiation mechanisms. Modification of the code for 2nd order accuracy in time marching is currently underway. The numerical simulations are being used as a predictive tool to assess the test matrix of the SRD for the space-based experiments. Only the cases which should reveal the phenomena will remain, along with some safety-factor redundancies. Due to time and g-jitter limitations, little of the planned test matrix can be adequately explored in the ground based test facilities, and the ACME definition will rely on the best effort in the modeling work. The experiments are meant to examine the relative importance of radiative emission and re-absorption, the adequacy of the detailed chemical kinetics mechanism and transport sub-models used, and the onset of flame front instabilities that could modify the extinction boundaries. Constant pressure and constant volume cases are also being compared.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Wu F, Jomaas G, Law CK. "An experimental investigation on self-acceleration of cellular spherical flames." Proceedings of the Combustion Institute. 2013. 34(1):937-45. https://doi.org/10.1016/j.proci.2012.05.068 , Jan-2013
Articles in Peer-reviewed Journals	Xin YX, Sung CJ, Law CK. "A mechanistic evaluation of Soret diffusion in heptane/air flames." Combustion and Flame, 2012 Jul;159(7):2345-51. http://dx.doi.org/10.1016/j.combustflame.2012.03.005 , Jul-2012